Probabilistic Ancestral Inference

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The Missing Link: Reconstructing Stochastic Data

Stochastic: involving random probability and unpredictability where future states cannot be precisely determined. In genetics, this manifests as missing or unobservable data — ancestors whose genotypes were never sequenced due to cost, sample degradation, or ethical constraints.

The Problem

Genetic data is rarely a perfect record. In ancestral genomics, sequencing every ancestor in a pedigree is often impossible due to cost, sample degradation, or ethical constraints. These gaps in family trees limit our ability to reconstruct inheritance patterns, predict hereditary traits, and model population history. If we simply ignore these gaps, our analysis becomes biased and our "family trees" of information fall apart.

The Objective

My research investigates whether probabilistic models can reconstruct missing ancestral genotypes by leveraging the surviving data points in the pedigree. By benchmarking five model architectures — Bayesian, HMM, DNN, GNN, and a frequentist baseline — against simulated ground truth, the goal is to identify which approaches are most reliable and where reconstruction breaks down as data becomes sparse.

The Genetics Sandbox: Why DNA?

The Test Lab: Generating "Perfect" History Simulating Ground Truth

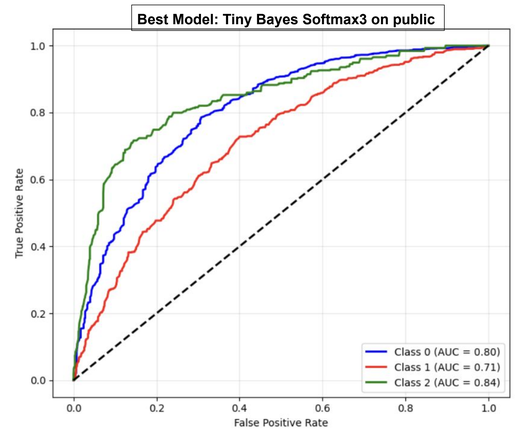
To evaluate whether my models actually work, I need a dataset where the ground truth is known — something impossible with real sequencing data. So I turned to a specialized Python library called msprime which allows me to simulate entire biological histories and ancestral lineages based on real world findings. While the data is simulated, msprime is validated in a peer-reviewed research published in the journal Genetics (Baumdicker et al., 2022) and used to simulate ancestral history and DNA sequence data.

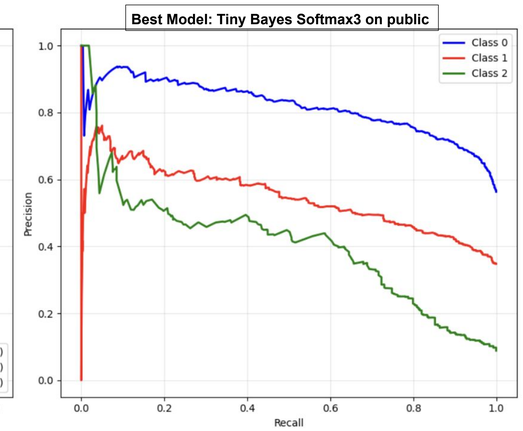
The Architecture: TypeScript & Python Integration

This project is built on a custom bridge between two worlds: React and Python to create a seamless application that leverages the strengths of both. React is a UI library made by Facebook (now Meta) for webpages while Python is the industry standard for AI and Machine Learning. I use FastAPI in Python to serve the webpage (React) with the data and necessary Python processing.

Probabilistic Modeling

The core of the project uses Probabilistic Models. Each model takes the features derived from an individual's observable relatives and outputs a probability distribution over the three possible dosage classes (0, 1, or 2). The predicted class is whichever dosage is most probable given the surrounding pedigree evidence.





Model Statistics

Model **tiny bayes_softmax3** was applied to dataset **public**.

Test Metrics — Stats for Nerds

Metric	Value	What it means
Accuracy	**62.92%**	Percentage of all genotype calls that were correct. Because most datasets are unbalanced — dosage 0 is far more common than 1 or 2 — a high accuracy can hide poor performance on rare genotypes.
Balanced Accuracy	**62.91%**	Like accuracy, but each genotype class is weighted equally regardless of how often it appears. Since datasets are usually unbalanced, this is a more honest measure of overall performance than plain accuracy.
Macro AUC	**78.57%**	How well the model separates each genotype class (0, 1, 2), averaged equally across all three. Unaffected by class imbalance. A score of 1.0 is perfect; 0.5 is no better than random.
F1 (Macro)	**54.29%**	Balance of precision and recall, averaged equally across all classes. Because datasets are unbalanced, this penalises the model heavily if it ignores a rare genotype class entirely.
F1 (Weighted)	**62.01%**	Same as F1 Macro, but classes that appear more often contribute more to the average. On unbalanced data this tends to look better than F1 Macro — use both together for a complete picture.

Model Statistics

Model **small multi_log_regression** was applied to dataset **public**.

Test Metrics

Metric	Value	What it means
Accuracy	**48.02%**	Percentage of all genotype calls that were dosage 0 is far more common than 1 or 2 genotypes.
Balanced Accuracy	**55.67%**	Like accuracy, but each genotype class is datasets are usually unbalanced, this is a accuracy.
Macro AUC	**73.68%**	How well the model separates each genot Unaffected by class imbalance. A score of
F1 (Macro)	**45.97%**	Balance of precision and recall, averaged unbalanced, this penalises the model hea
F1 (Weighted)	**50.41%**	Same as F1 Macro, but classes that appe unbalanced data this tends to look better t

Model	Accuracy	Bal. Accuracy	Macro-AUC	F1 Macro	F1 Weighted
tiny, bayes	0.6292	0.6291	0.7857	0.5429	0.6201
tiny, log_reg	0.6243	0.6161	0.7649	0.5337	0.6104
tiny, hmm	0.5527	0.5637	0.7559	0.4406	0.5375
small, dnn	0.521	0.5729	0.7197	0.4837	0.5452
tiny, dnn	0.5178	0.5481	0.7255	0.4667	0.5419
small, log_reg	0.4802	0.5567	0.7368	0.4597	0.5041
small, gnn	0.4224	0.474	0.6603	0.3942	0.453
tiny, gnn	0.4063	0.3523	0.5587	0.3399	0.4449
medium, log_reg	0.3861	0.5046	0.7077	0.382	0.3891
medium, dnn	0.3667	0.4966	0.6939	0.3488	0.3266
medium, gnn	0.3151	0.4746	0.6563	0.2875	0.2612
small, hmm	0.3929	0.4952	0.6858	0.3723	0.4321
medium, hmm	0.1692	0.3791	0.5982	0.1651	0.1377

If you want to check out the website: capstone.makennaworley.com (best done on computers/laptops)

LinkedIn: https://www.linkedin.com/in/makenna-worley/ | GitHub: https://github.com/MakennaWorley

What I Figured Out: Reconstructing the Past

Why isn't it 100%?

Reconstructing the past is a game of probability, not certainty. While our best models are 2x more accurate than guessing, they struggle as data becomes 'sparse.' Just as a detective can't solve a case with zero clues, our models lose accuracy when relatives are too far apart to provide a clear mathematical trail.

What I Figured Out

The most important takeaway is that the pedigree structure contains a real, exploitable signal — even when significant portions of ancestors are missing, the surrounding relatives provide enough context for models to reconstruct genotypes better than guessing. However, performance degrades sharply as dataset scale increases, pointing to a fundamental limit: the signal from relatives weakens as data gets sparser.

Obstacles

Because of the nature of stochastic data, the models can never be 100% accurate — and this showed clearly in the results. Accuracy dropped from ~63% on the tiny dataset to as low as ~17% (HMM) and ~37% (DNN, logistic regression) on the medium dataset. The hardest case to predict was always the heterozygous genotype (dosage = 1, one copy of each allele), because it's the most ambiguous under Mendelian inheritance. There is still significant research needed, but the initial results confirm that pedigree-based probabilistic inference is a viable approach.

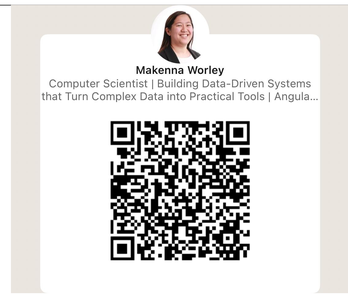
The Bayesian model achieved the highest accuracy but uses MCMC sampling, making it computationally expensive to scale — results at small and medium dataset sizes are still pending.

Beyond the Lab

Broad Applications

While I used genetics as my "controlled environment" for this project, the implications for this logic are much broader:

- **Historical Records:** Reconstructing damaged census data or ancient family lineages.
- **Ecology:** Reconstructing missing specimens in evolutionary phylogenetic trees.
- **Healthcare:** Filling in gaps in patient histories to better predict hereditary health risks.

Open Source Science

The logic and code for this reconstruction engine are available on my GitHub. The goal is to move toward models that can handle even larger, more complex stochastic datasets.